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DENVER RTD EAGLE P3

Communication System Software Test Plan

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1.0 Purpose

1.1 Introduction

This is the test procedure for the DENVER Communication System software. This procedure will address only those items and elements that are related to the DENVER Communication System, both directly and indirectly affected elements will be addressed. The primary focus of this procedure is to ensure that the DENVER Communication System provides DENVER Communication System operation by evaluating sets of features.

1.2 Scope

This test procedure covers a full software test of the DENVER Communication System.

2.0 Reference

The following documents are referenced

Publisher	Document Number	Title
DENVER	18FH012	DENVER RTD TS (Confirmed Version, March 2010) Part C Rolling Stock Requirements
Rotem	REDU 106113	Procurement Technical Specification For DENVER RTD EMU Communication System
WOOJIN	WJ-SRS-001_r2	DENVER RTD EAGLE P3 Communication System Software Requirements Specification
	WJ-SRS-SDD-V1.0	DENVER RTD EAGLE P3 Communication System Software Design Description

3.0 Test Item

All items that make up the DENVER Communication System software will be tested. The versions to be tested will be placed in the appropriate libraries by the configuration administrator. The administrator will also control changes to the versions under test and notify the test group when new versions are available.

The following documents will provide the basis for defining correct operation:

4.0 Features to be Tested

Refer to section 12.0

5.0 Features not to Tested

The test cases will not cover all possible combinations of options within the transaction or reports being tested. Only combinations that are known to be required for DENVER Communication System processing will be tested.

6.0 Approach

The software for LEDC and CNCS shall be integrated with corresponding hardware. The software for PAC, shall be connected with PCB using emulator.

And we will build a comprehensive set of test cases of input values and activities. The output value and result from the software/hardware integrated system will be compared with the expected output value in the user functional point of view.

Testing shall be performed once a complete software configuration item is embedded in hardware.
Routine test procedures developed during the design phases of the project shall be used to guide testing, and test reports shall be generated subsequently.

If any test procedure failure during formal type testing indicates a software error, then an anomaly shall be reported. Any failure during informal testing shall be handled as part of the normal product development cycle and is not recorded as an anomaly.

If any software error is identified during testing, Error correcting task shall be assigned to the software development group and re-submit the configuration items for re-testing.
If the identification of software problems results in modifications to software documentation, pertinent documents shall be modified and resubmitted.

7.0 Item Pass/Fail Criteria

Refer to section 12.0

8.0 Suspension Criteria and Resumption Requirements

8.1 Suspension Criteria

Inability to integrate with hardware(emulator) will cause suspension of all testing activities

8.2 Resumption Requirements

When a new version of the system is transmitted to the testing personnel after a suspension has occurred, a regression test will be run.

9.0 Test Deliverables

The following documents will be generated by the testing personnel and will be delivered to the configuration management personnel after test completion.

Software Test Report (including Test Execution Trace Report)

10.0 Environmental Needs

10.1 CCP

RS-485, CAB, Conductor and Address input signals should be connected and PAC, CNCS, FDI, SDI and PII devices also should be composed for CPP Software Test.

10.2 PAC

The result processed should be checked by connecting J-TAG debugger to PAC board and changing the software variable for the test lists for PAC Software test. Besides, It should be identified as RS485 input value by connecting the other device packet simulation program to RS485 and changing the input value.

- External Interface Signals Test : It should operate the input signal for the test lists and should be identified as RS485 input value by connecting External Interface Signals test board operating the Input and Output Port signals which are input to PAC board

- Multi coupled unit Process Test : The result value should be tested by using RS 485 and checked by connecting at least more than four PAC boards.

- Announcement Test : The result of announcement should be checked by connecting speakers.

- Polling Test : It should be tested by connecting the other device packet simulation program in the communication system to RS485.

10.3 LEDC

FDI, SDI, PII Unit with LEDC, PSA of CRA Rack, CNCS and PAC should be composed for LEDC Software test. Following the power approval, it should be checked that the device initial value and ID are set up as the front text of LED module(LDM). Besides, the normal operation status of FDI, SDI and PII Unit relevant should be identified by pressing the CHECK button of CCP.

10.4 LIC

PSA of CRA Rack, CNCS and PAC should be composed for LIC Software test. Following the power approval, it should be checked that the device initial value and ID are set up as the front text of LED module(LDM). Besides, the normal operation status of the unit relevant should be identified by pressing the CHECK button of CCP.

11.0 Responsibilities

	PM	Development Team	Testing Personnel
Create Test Cased		X	
Test Execution			X
Create Test Documents	X		

The development team leader will be responsible for creating test cases and test procedures. The project manager is responsible for relative documentation. The testing personnel is responsible for test execution.

The entire project team will participate in the review of the system and detail designs as well as review of any change requests that are generated by the user or as a result of defects discovered during development and testing.

12.0 Test Case Specifications

12.1 PAC

12.1.1 Fun-REQ-PAC-Initial-01

12.1.1.1 Special Requirements

None

12.1.1.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
PAC Initialization	1) Send CRA(MA) port signal from external interface single test board. 2) Send CRA(MB) port signal from external interface signal test board.	1) It is checked that CRA-AB bit becomes 1 in carline packet viewer. 2) It is checked that CRA-AB bit becomes 0 in carline packet viewer.	TC-PAC-INITIAL-01-01
	1) Change LIC packet in packet simulation program and send #4001 car numbers thru RS485. 2) Change LIC packet in packet simulation program and send #4002 car numbers thru RS485.	1) Check if CRA-MAB byte is 1 in carline packet viewer. 2) Check if CRA-MAB byte is 2 in carline packet viewer.	TC-PAC-INITIAL-01-02
	1) Send CAB ON port signal from external interface signal test board. 2) Send CAB OFF port signal from external interface signal test board.	1) It is checked that CAB bit is 1 in carline packet viewer. 2) It is checked that CAB bit is 0 in carline packet viewer.	TC-PAC-INITIAL-01-03
	1) Send coupler ON port signal from external interface signal test board. 2) Send coupler OFF port signal from external interface signal test board.	1) Check if coupler bit 1 in carline packet viewer. 2) Check if coupler bit 0 in carline packet viewer.	TC-PAC-INITIAL-01-04
	1) Send CRA(MA) port signal of external interface signal test board on car #4001. 2) Send CRA(MA) port signal of external interface signal test board on car #4003.	1) It works as a main PAC of car and polling packets as many as car device IDs(34) are observed. 2) It works as a main PAC of car and polling packets as many as car device IDs(34) are observed.	TC-PAC-INITIAL-01-05
	1) Set PAC internal program variable to #4001 and #4003 to double headed train. 2) Send CRA(MA) port single	1) It is checked that temporary change of internal variable in carline packet viewer. 2) Carline polling runs by main	TC-PAC-INITIAL-01-06

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
	from external interface single test board. 3) Send CAB on port signal from external interface signal test board.	PAC. Carline polling of #4001, #4003 train runs by master PAC.	
	1) Change LIC packet in packet simulation program and send #4005 car number thru RS485. 2) Off the packet simulation. 3) Reboot PAC board.	1) Save the car number #4005 from LIC packet on PAC flash memory and RAM. 2) Runs as car number which save in RAM. 3) LIC's car number existing in the absence of information, information work and are loaded from flash memory.	TC-PAC-INITIAL-01-07
	1) PAC power ON. 2) Change LIC packet in packet simulation program and send #4007 car number thru RS485.	1) It runs by loading car number information from internal flash memory. 2) Save the car number from LIC packet on PAC flash memory and RAM and runs #4007.	TC-PAC-INITIAL-01-08
	1) Power ON of CRA(MA) PAC and CRA(MB) PAC. 2) Power OFF CRA(MA) PAC.	1) CRA(MA) PAC is polled to main PAC. 2) Detect if CRA(MB) PAC and CRA(MA) PAC are dead and are running by changing the role from sub PAC to main PAC.	TC-PAC-INITIAL-01-09

12.1.1.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.1.2 Fun-REQ-PAC-Abalysis-01

12.1.2.1 Special Requirements

None

12.1.2.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
Poll and record the status of all RS-485 Car Line devices	1) Send CRA(MA) port signal from external interface signal test board. 2) Power on PAC board.	1) Run as main PAC. 2) Polling continuously as many as device IDs.	TC-PAC-ANALYSIS-01-01
	1) Make all other device packets with packet simulation program and send via RS485. 2) Cancel the packet response of some devices.	1) If the device gives a feedback of response packet to polling packet of PAC, the device is displayed alive on the status information of communication rack. 2) If there were no response of packet received, the device that didn't receive the packet would be displayed dead.	TC-PAC-ANALYSIS-01-02

12.1.2.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.1.3 Fun-REQ-PAC-Analysis-02

12.1.3.1 Special Requirements

None

12.1.3.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
Consist Inauguration	1) Turn on CRA(MA) and CAB port signal of external interface signal test board in car #4001. 2) Turn on CRA(MB) and CAB port signal of external interface signal test board in car #4002.	1) It is checked with carline packet viewer that CAB is on at CRA(MA) on car #4001 and train is connected with #4001, #4002, #4003 and #4004. 2) It is checked with carline packet viewer that CAB is on at CRA(MB) in car #4002 and train is connected with #4004, #4003, #4002 and #4001.	TC-PAC-ANALYSIS-02-01
	1) Turn on CRA(MA) and CAB port signal of external interface signal test board in car #4003. 2) Turn in CRA(MB) and CAB port signal of external signal test board in car #4004.	1) It is checked with carline packet viewer that CAB is on at CRA(MA) on car #4003 and train is connected with #4001, #4002, #4003 and #4004. 2) It is checked with carline packet viewer that CAB is on at CRA(MB) on car #4004 and train in connected with #4004, #4003, #4002 and #4001.	TC-PAC-ANALYSIS-02-02

12.1.3.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.1.4 Fun-REQ-PAC-Analysis-03

12.1.4.1 Special Requirements

None

12.1.4.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
Retrieve the Device Status tables from all PRIMARY PAC's	1) Turn on CRA(MA) and CAB port signal of external interface signal test board in car #4001. 2) Turn on CRA(MB) and CAB port signal of external interface signal test board in car #4002.	1) Device status table is created because CAB is on at CRA(MA) on car #4001 and double heading is complete. 2) Device status table has same configuration as CAB on at CRA(MA) even if CAB is on at CRA(MB) in car #4002.	TC-PAC-ANALYSIS-03-01
	1) Create CCP packet that crew ID and train number are changed on packet simulation program of car #4001.	1) Check if the information of master PAC with CAB on is reviewed with the same train number and crew ID on car #4003.	TC-PAC-ANALYSIS-03-02
	1) #4003 CRA(MA), #4004 CRA(MB) PAC Power OFF.	1) Check if the dead device information of car #4003 is renewed on master PAC #4001.	TC-PAC-ANALYSIS-03-03

12.1.4.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.1.5 Fun-REQ-PAC-Analysis-04

12.1.5.1 Special Requirements

None

12.1.5.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
PA Audio Train Line level balancing	1) Turn on CRA(MA) and CAB port signal of external interface signal test board in car #4001.	1) During double heading, 11Khz tone occurred is viewed on train line packet viewer.	TC-PAC-ANALYSIS-04-01
	1) Change the resistance of volume gain on external interface signal test board. 2) Load the program which volume gain can't be adjusted with respect to voltage values. 3) Reboot PAC board. 4) Load the program which volume gain can be adjusted with respect to voltage values. 5) Change the resistance of volume gain on external interface signal test board. 6) Reboot PAC board.	1) Load on audio line of train line. 2) Test the program which doesn't balance the level before the test. 3) When double heading is complete, check that CNCS default volume and PAC Amp volume are the same. 4) Load the final version of program to test. 5) Load on audio line of train line. 6) After double heading and level balance applied, check that the PAC Amp volume is increased on carline packet viewer.	TC-PAC-ANALYSIS-04-02

12.1.5.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.1.6 Fun-REQ-PAC-Analysis-05

12.1.6.1 Special Requirements

None

12.1.6.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
Monitor Consist Integrity	1) Turn on CRA(MA) and CAB port signal of external interface signal test board in car #4001. 2) Turn off CAB port signal of external interface signal test board in car #4001.	1) It's shown in carline packet viewer that CAB is on at CAR in car #4001 and if train is connected with #4001, #4002, #4003 and #4004. 2) The information of double headed train is set to initial value 10 sec after CAB signal is off which is viewed on carline packet viewer.	TC-PAC-ANALYSIS-05-01
	1) Turn on CAR and CAB port signal of external interface signal test board in car #4001. 2) Change coupler port signal of external interface signal test board on TOC carline #4002.	1) It's shown in carline packet viewer that CAB is on at CRA(MA) on car #4001 and if train is connected with #4001, #4002, #4003 and #4004. 2) On double heading, if coupler signal of HOC or TOC is changed, then it's shown in train line packet viewer that re-double heading begins.	TC-PAC-ANALYSIS-05-02

12.1.6.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.1.7 Fun-REQ-PAC-Analysis-06

12.1.7.1 Special Requirements

None

12.1.7.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
Monitor Intercom Status	1) Turn in DPO live signal of external interface signal test board. 2) Turn off DPO live signal of external signal test board.	1) It is checked on carline packet viewer that DPO live bit of device status tables is 0. 2) It is checked on carline packet viewer that DPO live bit of device status tables is 1.	TC-PAC-ANALYSIS-06-01
	1) Turn on DPH live signal of external interface signal test board. 2) Turn off DPH live signal of external interface signal test board.	1) It is checked on carline packet viewer that DPH live bit of device status tables is 0. 2) It is checked on carline packet viewer that DPH live bit of device status tables is 1.	TC-PAC-ANALYSIS-06-02
	1) Turn on PEI live signal of external interface signal test board. 2) Turn off PEI live signal of external interface signal test board.	1) It is checked on carline packet viewer that PEI live bit of device status tables is 0. 2) It is checked on carline packet viewer that PEI live bit of device status tables is 1.	TC-PAC-ANALYSIS-06-03

12.1.7.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.1.8 Fun-REQ-PAC-Address-01

12.1.8.1 Special Requirements

None

12.1.8.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
Process Local PEI Address Status Interrupt	1) Turn on PEI address signal of external interface signal test board. 2) Turn off PEI address signal of external signal test board.	1) It is checked on carline packet viewer that PEI address bit of device status tables is 1. 2) It is checked on carline packet viewer that PEI address bit of device status tables is 0.	TC-PAC-ADDRESS-01-01

12.1.8.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.1.9 Fun-REQ-PAC-Address-02

12.1.9.1 Special Requirements

None

12.1.9.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
Query Train Line PEI Address Status	1) Turn on CRA(MA) and CAB port signal of external interface signal test board. 2) Turn on PEI address signal of external interface signal test board.	1) It's shown in carline packet viewer that CAB is on at CAR on car #4001 and train is connected with #4001, #4002, #4003 and #4004. 2) It's shown in carline packet viewer that PEI address information called form car #4001 is renewed on car #4003.	TC-PAC-ADDRESS-02-01
	1) Turn on PEI address signal of external interface signals test board at car #4003 or #4004.	1) It is checked in carline packet viewer that PEI address information called from car #4003 or #4004 is renewed on master PAC #4001.	TC-PAC-ADDRESS-02-02

12.1.9.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.1.10 Fun-REQ-PAC-Initial AVL-01

12.1.10.1 **Special Requirements**

None

12.1.10.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
Monitor & Process AVL Trigger Events	1) Turn on CRA(MA) and CAB port signal of external interface signal test board in car #4001. 2) Create CCP packet which the trigger point is change on the packet simulation program of car #4001 and send it via RS485.	1) It's shown in carline packet viewer that CAB is on at CRA(MA) on car #4001 and train is connected with #4001, #4002, #4003 and #4004. 2) Trigger point information recognized at master PAC in car #4001 is also renewed in car #4003.	TC-PAC-INITAVL-01-01

12.1.10.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.2 CCP

12.2.1 Fun-REQ-CCP-Initial-01

12.2.1.1 **Special Requirements**

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.1.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Approve the power to the system.	1. Booting by inputting a logo on the screen then CCP program is operated.	TC-CCPInitail-01
	2. Approve the signals.	2."Press Train to enter Train Number". A string is displayed.	

12.2.1.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.2.2 Fun-REQ-CCP Control-01

12.2.2.1 Special Requirements

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.2.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Input the Train number valid	1. Load the driving information for Train Number in the memory and display on the screen.	TC-CCPControl-01
	2. Input the Train number invalid	2. Display ”Invalid Train Number”	

12.2.2.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.2.3 Fun-REQ-CCP Control-02

12.2.3.1 Special Requirements

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.3.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Input CREW ID on the input-screen.	1. CREW ID is loaded and "In Service" is displayed	TC-CCPControl-02

12.2.3.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.2.4 Fun-REQ-CCP Control-03

12.2.4.1 Special Requirements

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.4.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Input the Train number valid.	1. The Train Number which is input in PAC and the communication packet is transmitted by applying.	TC-CCPControl-03
	2. Press the SKIP button of a specific station on the screen.	2. No automatic announcement for the skip button applied station when a simulation is operated.	
	3. Press the EXP button of a specific station on the screen.	3. No automatic announcement for the EXP button applied the section of a station when a simulation is operated.	
	4. Press the STOP button of a specific station on the screen.	4. Automatic announcement for the STOP button applied station when a simulation is operated.	

12.2.4.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.2.5 Fun-REQ-CCP Control-04

12.2.5.1 **Special Requirements**

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.5.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Press the NORMAL button on the CONFIG-SPEAKER VOLUME screen.	1. The speaker volume is adjusted by the normal level when automatic announcement.	TC-CCPControl-04
	2. Press the SOFT button on the CONFIG-SPEAKER VOLUME screen.	2. The speaker volume is adjusted by the SOFT level when automatic announcement.	
	3. Press the MUTE button on the CONFIG-SPEAKER VOLUME screen.	3. No sounds are input when automatic announcement.	
	4. Adjust a scroll bar on the CONFIG – LCD DIMMER screen.	4. The brightness of the screen is adjusted by the direction of the scroll bar.	
	5. After change of speaker volume and screen brightness, press the RESET button.	5. After the reoperation of the system the screen brightness value is set as the initial value.	

12.2.5.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.2.6 Fun-REQ-CCP Control-05

12.2.6.1 Special Requirements

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.6.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Check the CHECK screen after forming a single car.	1. Only one car is shown as an icon. The rest of the car area is filled with white empty space.	TC-CCPControl-05
	2. Check the CHECK screen after the forming a multi car.	2. The number of the composed cars is shown as icons. The rest of the car area is filled with white empty space.	
	3. Remove the power of the specific car formation device.	4. The icon color of the car with relevant devices turns red. If click a car icon, the device removed the power becomes red.	

12.2.6.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.2.7 Fun-REQ-CCP Control-06

12.2.7.1 Special Requirements

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.7.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Click the lists of the screen one by one.	1. The relevant information of the lists chosen is displayed.	TC-CCPControl-06

12.2.7.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.2.8 Fun-REQ-CCP Control-07

12.2.8.1 **Special Requirements**

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.8.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Press the PEI CALL button in a specific car after forming a multi car.	1. The PEI button color is changed when the message including the car number with the button pushed on the CCP screen is displayed	TC-CCPControl-07

12.2.8.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.2.9 Fun-REQ-CCP Control-08

12.2.9.1 **Special Requirements**

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.9.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Press the PEI CALL button in a specific car after forming a multi car.	1. The PEI button color is changed when the message including the car number with the button pushed on the CCP screen is displayed	TC-CCPControl-08

12.2.9.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.2.10 **Fun-REQ-CCP Control-10**

12.2.10.1 **Special Requirements**

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.10.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Update the DB file by using CNCS Content Update program.	1. The file updated from CNCS is stored in CCP storage device.	TC-CCPControl-10

12.2.10.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.2.11 **Fun-REQ-CCP Control-11**

12.2.11.1 **Special Requirements**

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.11.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Update the DB file by using CNCS Content Update program.	1. The file updated from CNCS is stored in CCP storage device.	TC-CCPControl-11

12.2.11.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.2.12 **Fun-REQ-CCP Control-12**

12.2.12.1 **Special Requirements**

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.12.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Update the DB file by using CNCS Content Update program.	1. The file updated from CNCS is stored in CCP storage device.	TC-CCPControl-12

12.2.12.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.2.13 **Fun-REQ-CCP Display-01**

12.2.13.1 **Special Requirements**

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.13.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Input the normal Train Number.	1. Departure, Actual and next station names for the Train number input by the main user on the main screen are displayed.	TC-CCPDisplay-01

12.2.13.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.2.14 **Fun-REQ-CCP Comm-02**

12.2.14.1 **Special Requirements**

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.14.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Input the Train Number and execute the Simulation.	1. Automatic announcement is output by the schedule. 2. Check If Actual stations on CNCS screen are changed by a schedule.	TC-CCPComm-02
	1. Input the Train Number and transmit GIA packet to CCP by checking the TP information of the schedule input.	1. Automatic announcement is output by the location of GIA packet transmitted.	

12.2.14.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.2.15 **Fun-REQ-CCP Comm-03**

12.2.15.1 **Special Requirements**

Not only should RS-485 communication lines, CAB, Conductor and Address input signals be connected but PAC, CNCS, FDI, SDI and PII devices also should be composed for CCP software test.

12.2.15.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CCP Interface	1. Input the Train Number and transmit GIA packet to CCP by checking the TP information of the schedule input.	1. Automatic announcement is output by the location of GIA packet transmitted and the station information of the main screen is displayed with a change.	TC-CCPComm-03

12.2.15.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.3 CNCS

12.3.1 Fun-REQ-CNCS Launcher-01

12.3.1.1 Special Requirements

PAC in CRA Rack, CNCS, CCP, LED and PID Display should be composed for CNCS Software test.

12.3.1.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	1) Add a normal update file in the update folder. 2) Add an abnormal update file in the update folder.	1) In case of normal performance, the file is updated. 2) In case of normal performance, the file is not updated.	TC-CNCS Launcher-01

12.3.1.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.3.2 Fun-REQ-CNCS Main-Initial-01

12.3.2.1 Special Requirements

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.2.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	1) In case of normal performance, CNCS will ordinarily respond to the requirement of PAC. 2) In case of normal performance, Sub-Component should be monitored.	1) CNCS is displayed as “OK” on the right side of CCP. 2) Process Monitor State of the Main Screen shows the state of Sub-Component.	TC-CNCS Main-Initial-01

12.3.2.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.3.3 Fun-REQ-CNCS Main-Monitoring-01

12.3.3.1 Special Requirements

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.3.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	1) After 5 minutes, processor should ordinarily be operated.	1) The processor operates ordinarily after 5 minutes.	TC-CNCS Main-Monitoring-01

12.3.3.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.3.4 Fun-REQ-CNCS Main-RS485-01

12.3.4.1 Special Requirements

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.4.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	1) In case of normal performance, CNCS will ordinarily respond to the requirement of PAC.	1) CNCS is displayed as “OK” on the right side of CCP.	TC-CNCS Main-RS485-01

12.3.4.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.3.5 Fun-REQ-CNCS PP-Initial-01

12.3.5.1 Special Requirements

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.5.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	1. Switch on CNCS and wait until the contents play back by operating PP.	1. Check if an advertisement video or a still image normally play on the LCD Display screen and confirm if the PP state is 2 when stop.	TC-CNCS PP-Initial-01

12.3.5.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.3.6 Fun-REQ-CNCS PP-ContentPlay-01

12.3.6.1 Special Requirements

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.6.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	1. Add the driving information(car and crew numbers) in CCP and test it by using simulation functions.	1. Observe If the passenger guide information is displayed by the schedule with car numbers.	TC-CNCS- PP-ContentPlay-01

12.3.6.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.3.7 Fun-REQ-CNCS SP-Initial-01

12.3.7.1 Special Requirements

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.7.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	1. Switch on CNCS and wait until UDP communication is on by using SP.	1. Check if SP state of Main.exe is 2.	TC-CNCS SP-Initial-01

12.3.7.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.3.8 Fun-REQ-CNCS SP-ContentPlay-01

12.3.8.1 Special Requirements

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.8.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	1. Switch on CNCS and wait until UDP communication is on by using SP.	1. Check if SP state of Main.exe is 2.	TC-CNCS SP-ContentPlay-01

12.3.8.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.3.9 Fun-REQ-CNCS MDS Data-Initial-01

12.3.9.1 Special Requirements

PAC in CRA Rack, CNCS, CCP, LED and PID Display should be composed for CNCS Software test.

12.3.9.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	LIC response should be confirmed on the MDS Data screen.	The response is conformed on the MDS Data screen.	TC-CNCS MDS Data-Initial-01

12.3.9.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.3.10 Fun-REQ-CNCS MDS Data-RS232-01

12.3.10.1 **Special Requirements**

PAC in CRA Rack, CNCS, CCP, LED and PID Display should be composed for CNCS Software test.

12.3.10.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	The MDS data file from LIC should be created.	The MDS data file from LIC was created.	TC-CNCS MDS Data-RS232-01

12.3.10.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.3.11 Fun-REQ-CNCS MDS Data-RS232-02

12.3.11.1 **Special Requirements**

PAC in CRA Rack, CNCS, CCP, LED and PID Display should be composed for CNCS Software test.

12.3.11.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	The time information of CNCS should be transmitted to LIC.	The time information should be confirmed on the MDS Data Screen.	TC-CNCS MDS Data-RS232-02

12.3.11.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.3.12 **Fun-REQ-CNCS MDS Data-RS232-03**

12.3.12.1 **Special Requirements**

PAC in CRA Rack, CNCS, CCP, LED and PID Display should be composed for CNCS Software test.

12.3.12.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	The version information of CNCS should be transmitted to LIC.	The version information should be confirmed on the MDS Data Screen.	TC-CNCS MDS Data-RS232-03

12.3.12.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.3.13 **Fun-REQ-CNCS FTP-Initial-01**

12.3.13.1 **Special Requirements**

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.13.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	The normal state of FTP should be displayed on the Main Process.	The normal state of FTP was displayed on the Main Process.	TC-CNCS FTP-Initial-01

12.3.13.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.3.14 **Fun-REQ-CNCS FTP-Wayside-01**

12.3.14.1 **Special Requirements**

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.14.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	If FTP Server is normal, the Wayside Icon of CCP should be displayed with green color.	When FTP Server is normal, the Wayside Icon of CCP was displayed with green color.	TC-CNCS FTP-Wayside-01

12.3.14.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.3.15 Fun- REQ-CNCS FTP-Update-01

12.3.15.1 **Special Requirements**

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.15.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	If the target file is transmitted normally, two file have the same contents.	The target file and uploaded file into the FTP Server are same.	TC- REQ-CNCS FTP-Update-01

12.3.15.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.3.16 Fun-REQ-CNCS FTP-Update-02

12.3.16.1 **Special Requirements**

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.16.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	The target file and downloaded file into the CNCS should be same.	The target file and downloaded file into the CNCS are same.	TC-CNCS FTP-Update-02

12.3.16.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.3.17 Fun-REQ-CNCS ContentUpdate-Initial-01

12.3.17.1 Special Requirements

PAC in CRA Rack, CNCS, CCP, LED and PID Display should be composed for the test of CNCS Software.

12.3.17.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	1. The new data of CCP should be normally identified in Content Update Process	1. The new data of CCP was normally identified in Content Update Process.	TC-CNCS ContentUpdate-Update-01
	2. The new data of PAC should be normally identified in Content Update Process	1. The new data of PAC was normally identified in Content Update Process.	
	3. The new data of LED should be normally identified in Content Update Process	1. The new data of LED was normally identified in Content Update Process.	

12.3.17.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.3.18 Fun-REQ-CNCS ContentUpdate-RS485-01

12.3.18.1 Special Requirements

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.18.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	Data should ordinarily be transmitted and received via Car Line.	RS-485 transmitting and receiving states were displayed normally.	TC-CNCS ContentUpdate-RS485-01

12.3.18.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.3.19 Fun-REQ-CNCS ContentUpdate-Update-01

12.3.19.1 Special Requirements

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.19.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	New data of CCP should be recognized normally in the Content Update Process.	New data of CCP was recognized normally in the Content Update Process.	TC-CNCS ContentUpdate-Update-01
	New data of PAC should be recognized normally in the Content Update Process.	New data of PAC was recognized normally in the Content Update Process.	
	New data of LED should be recognized normally in the Content Update Process.	New data of LED was recognized normally in the Content Update Process.	

12.3.19.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.3.20 Fun-REQ-CNCS ContentUpdate-Update-02

12.3.20.1 Special Requirements

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.20.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	If simulation is performed on the CCP, the updated data should be shown.	The updated data is shown normally.	TC-CNCS ContentUpdate-Update-02

12.3.20.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.3.21 Fun-REQ-CNCS ContentUpdate-Update-03

12.3.21.1 **Special Requirements**

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.21.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	If simulation is performed on the CCP, the updated data should be announced.	The updated data is announced normally.	TC-CNCS ContentUpdate-Update-03

12.3.21.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.3.22 Fun-REQ-CNCS ContentUpdate-Update-04

12.3.22.1 **Special Requirements**

The CNCS and PAC in CRA Rack and CCP, LED and PID Display are required in order to perform the CNCS Software test.

12.3.22.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	If simulation is performed on the CCP, the updated data should be shown.	The updated data is normally shown on the LED screen.	TC-CNCS ContentUpdate-Update-04

12.3.22.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.4 LEDC

12.4.1 Fun-REQ-FDI-Initial-01

12.4.1.1 Special Requirements

None

12.4.1.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
LEDC Signal Interface	After provide the power, set the initial value and ID of device and open the RS-485 port.	In case of normal operation, the LED of LEDC turns on.	TC-FDI-Initial-01

12.4.1.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.4.2 Fun-REQ-FDI-Comm-01

12.4.2.1 Special Requirements

FDI Unit with LEDC, CCP, PSA in CRA Rack, CNCS and PAC should be composed for the test of LEDC Software.

12.4.2.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
LEDC Signal Interface	Push the CHECK button of CCP.	In case of normal operation, the relevant FDI Unit indicates as green color, on the other hand, it indicates an red.	TC-FDI-Comm-01

12.4.2.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.4.3 Fun-REQ-FDI-Display-01

12.4.3.1 Special Requirements

FDI Unit with LEDC, CCP, PSA in CRA Rack, CNCS and PAC should be composed for the test of LEDC Software.

12.4.3.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
LEDC Signal Interface	After input CREW ID and train number using CCP GUI, push the simulation button.	In case of normal operation, the destination which is input, is displayed on the LED module(LDM) of the relevant FDI Unit	TC-FDI-Display-01

12.4.3.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.4.4 Fun-REQ-FDI-Update-01

12.4.4.1 Special Requirements

FDI Unit with LEDC, CCP, PSA in CRA Rack, CNCS and PAC should be composed for the test of LEDC Software.

12.4.4.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
LEDC Signal Interface	After copy a new image in Update folder of CNCS, operate the Update command. And after input CREW ID and train number using CCP GUI, push the simulation button.	In case of normal operation, a new destination image which was updated, is displayed on the LED module(LDM) of the relevant FDI Unit	TC-FDI-Update-01

12.4.4.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.4.5 Fun-REQ-SDI-Initial-01

12.4.5.1 Special Requirements

None

12.4.5.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
LEDC Signal Interface	After provide the power, set the initial value and ID of device and open the RS-485 port.	In case of normal operation, the LED of LEDC turns on.	TC-SDI-Initial-01

12.4.5.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.4.6 Fun-REQ-SDI-Comm-01

12.4.6.1 Special Requirements

SDI Unit with LEDC, CCP, PSA of CRA Rack, CNCS and PAC should be composed for LEDC Software test.

12.4.6.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
LEDC Signal Interface	Push the CHECK button of CCP.	In case of normal operation, the relevant FDI Unit indicates as green color, on the other hand, it indicates an red.	TC-SDI-Comm-01

12.4.6.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.4.7 Fun-REQ-SDI-Display-01

12.4.7.1 Special Requirements

SDI Unit with LEDC, CCP, PSA of CRA Rack, CNCS and PAC should be composed for LEDC Software test.

12.4.7.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
LEDC Signal Interface	After input CREW ID and train number using CCP GUI, push the simulation button.	In case of normal operation, the destination which is input, is displayed on the LED module(LDM) of the relevant FDI Unit	TC-SDI-Display-01

12.4.7.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.4.8 **Fun-REQ-SDI-Update-01**

12.4.8.1 **Special Requirements**

SDI Unit with LEDC, CCP, PSA of CRA Rack, CNCS and PAC should be composed for LEDC Software test.

12.4.8.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
LEDC Signal Interface	After copy a new image in Update folder of CNCS, operate the Update command. And after input CREW ID and train number using CCP GUI, push the simulation button.	In case of normal operation, a new destination image which was updated, is displayed on the LED module(LDM) of the relevant FDI Unit	TC-SDI-Update-01

12.4.8.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.4.9 Fun-REQ-PII-Initial-01

12.4.9.1 Special Requirements

None

12.4.9.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
LEDC Signal Interface	After provide the power, set the initial value and ID of device and open the RS-485 port.	In case of normal operation, the LED of LEDC turns on.	TC-PII-Initial-01

12.4.9.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.4.10 Fun-REQ-PII-Comm-01

12.4.10.1 Special Requirements

PII Unit with LEDC, CCP, PSA of CRA Rack, CNCS and PAC should be composed for LEDC Software test.

12.4.10.2 Features to be Tested

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
LEDC Signal Interface	Push the CHECK button of CCP.	In case of normal operation, the relevant FDI Unit indicates as green color, on the other hand, it indicates an red.	TC-PII-Comm-01

12.4.10.3 Pass/Fail Criteria

Each feature must pass all of its test cases in order to pass this test.

12.4.11 **Fun-REQ-PII-Display-01**

12.4.11.1 **Special Requirements**

PII Unit with LEDC, CCP, PSA of CRA Rack, CNCS and PAC should be composed for LEDC Software test.

12.4.11.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
LEDC Signal Interface	After input CREW ID and train number using CCP GUI, push the simulation button.	In case of normal operation, a text message which is input, is displayed on the LED module(LDM) of the relevant FDI Unit	TC-PII-Display-01

12.4.11.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.4.12 **Fun-REQ-PII-Update-01**

12.4.12.1 **Special Requirements**

PII Unit with LEDC, CCP, PSA of CRA Rack, CNCS and PAC should be composed for LEDC Software test.

12.4.12.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
LEDC Signal Interface	After copy a new image in Update folder of CNCS, operate the Update command. And after input CREW ID and train number using CCP GUI, push the simulation button.	In case of normal operation, a new destination image which was updated, is displayed on the LED module(LDM) of the relevant FDI Unit	TC-PII-Update-01

12.4.12.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.

12.5 LIC

12.5.1 Fun-REQ-LIC-Main-Comm-01

12.5.1.1 **Special Requirements**

PAC in CRA Rack, CNCS and CCP should be composed for the test of LIC Software.

12.5.1.2 **Features to be Tested**

Features to be Tested	Tests		Test Case ID
	Test Step	Expected Output	
CNCS Interface	1. After connecting RS-232 serial port the booting procedure can be checked by using 232 monitoring program.	1. The device name and the version are displayed when a normal operation. Ex1)[ENTER] : Move next line after out '->' Ex1)'?': Help Ex2)'T' : Date&time setting ,'TYYMMDDHHMNSS', Ex3)'t' : Date&time read Ex4)'V' : Version Ex5)'Recent' : Recently faults start Ex6)'Historical' : Historical data start Ex7)'Rxinfo' : Receive info Ex8)'LnWayClr' : 'm_nLnWkWaySideOnRxOk' variable clear Ex9)'LnFtpClr' : 'm_bLnWkFtpEndFlag' variable clear Ex10)'HisStTm' : Historical data start	TC-LIC Main-Comm-01

12.5.1.3 **Pass/Fail Criteria**

Each feature must pass all of its test cases in order to pass this test.